

28	D	$\Delta E = [hc/(60 \times 10^{-9}) - hc/(70 \times 10^{-9})] / (1.6 \times 10^{-19}) = 3 \text{ keV}$
29	B	Isotopes have the same number of protons but different number of neutrons. By emitting one α and two β particles, the number of protons of the element will be conserved (while the number of neutrons changes).
30	A	Initially $D = 0$, $C = 1000$